

Code-Layout, Readability and Reusability

Date	2 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the SmartLender project's code quality in terms of layout, readability, maintainability, and reusability. The project follows a modular structure with separate folders for templates, static files, machine learning models, and Flask backend code. Proper naming conventions, comments, and reusable functions make the application easy to understand, maintain, and extend.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Follows Python and HTML coding standards
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, models, and backend
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Easy to understand and maintain
4	Function / Method Names	Functions are descriptively named	Yes	Functions perform specific tasks
5	Code Comments	Inline and block comments explain logic	Yes	Improves code readability
6	Modular Design	Code is split into reusable functions/modules	Yes	Flask routes and ML model separated
7	No Redundant Code	Duplicate or unused code is removed	Yes	Optimized project structure
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Basic exception handling implemented

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
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1	Flask Application (app.py)	Python, Flask	Backend Request Handling	High
2	XGBoost Prediction Model (.pk1)	Python, XGBoost	Loan Predictio	High
3	HTML Templates	HTML, CSS, JavaScript	User Interface	High
4	Data Preprocessing Functions	Python, Pandas	Model Training & Prediction	Medium
5	input Validation Logic	Python, Flask	Form Processing	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure
Readability	5	Clear naming conventions and formatting
Reusability	5	Modular code with reusable components
Documentation / Comments	4	Important functions are documented
Overall Score	4.8 / 5	Clean, maintainable, and reusable codebase suitable for deployment